

Day 2 Lab Sheet: Install Tools, Use the Terminal, Make the Notes Board Work

Goal: By the end of this lab you will have Node.js and Git installed, you will have used the terminal, and your Notes Board will add and delete notes in the browser.

Time: about 100 minutes, plus the quiz.

What you need: The `Workshop-Installers` folder from your Day 1 homework, and your `Workshop` folder with the Day 1 result.

If your Day 1 files are missing or broken, copy the **Day 2 starter** folder from your instructor. It has everything finished up to Day 1. The **done** folder is the finished result for today.

Part 1: Install Node.js (10 minutes)

1. Open your `Workshop-Installers` folder and find the **Node.js 22 LTS** installer. It is a file that ends in `.msi`.
2. **Double-click** it. If Windows asks "Do you want to allow this app to make changes?", click **Yes**.
3. In the installer, click **Next** on each screen and keep the default choices. Accept the licence agreement when asked.
4. If you see a page about **"Tools for Native Modules"**, do **not** tick the box.
5. Click **Install**, wait, and click **Finish**.

Do not open VS Code yet. If VS Code was already open, you will close it in Part 3.

Part 2: Install Git (10 minutes)

1. In your `Workshop-Installers` folder, find the **Git for Windows** installer. It is a file that ends in `.exe`.
 2. **Double-click** it. Click **Yes** if Windows asks for permission.
 3. Click **Next** on each screen and keep the default choices.
 4. If you see a screen called **"Choosing the default editor used by Git"**:
 - Choose **Visual Studio Code** if it is in the list.
 - Otherwise choose **Notepad**.
 - Do **not** choose Vim. It is hard for beginners to close.
 5. Click **Install**, wait, and click **Finish**.
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Part 3: Restart VS Code and open the terminal (5 minutes)

A terminal learns about new programs only when it starts. So we restart VS Code first.

1. **Close VS Code completely** (click the X on the window).
2. Open VS Code again. It should open your `Workshop` folder. If not, click **File**, then **Open Recent**.
3. Click **Terminal** in the top menu, then **New Terminal**.
4. In the terminal panel, click the small **down arrow** next to the **+** icon, and choose **Command Prompt**.

Check: At the bottom you see a prompt like `C:\Users\YourName\Documents\Workshop>`.

Part 4: Check your tools (5 minutes)

Type each command and press **Enter**. After each one, compare what you see.

```
node -v
```

You should see a version that starts with `v22`, for example `v22.x.x`.

```
npm -v
```

You should see a version number.

```
git --version
```

You should see something like `git version 2.x.x`.

If you see "is not recognized as an internal or external command": close VS Code, open it again, open a new terminal, and try again. If it still fails, see the Stuck Sheet.

Run the setup check

1. Find the file `setup-check.js` in the installer kit folder (your instructor will tell you where it is on the LAN share or USB).
2. Copy it into your `Workshop` folder.
3. In the terminal, type:

```
node setup-check.js
```

Check: All four lines show **PASS**. Do not move on until they do. Raise your hand if a line shows **FAIL**.

Part 5: Practice with the terminal (10 minutes)

Type these commands one by one. Read what each one prints.

```
cd
```

This shows where you are.

```
dir
```

This lists the files and folders. You should see `notes-board` .

```
mkdir practice  
cd practice
```

You made a new folder and went inside it. Look at the prompt. It changed.

```
echo Hello from the terminal > hello.txt  
type hello.txt
```

The first line wrote a file. The second line showed what is inside it.

```
cd ..
```

You went one folder up. Look at the prompt again.

Now try these useful tricks:

- Type `cd note` and press **Tab**. The terminal completes the name `notes-board` .
- Press the **Up arrow**. Your last command comes back.
- Type `cls` and press Enter. The screen clears.

Check: You can move into a folder and out of it, and you can explain what the prompt shows.

Part 6: Try JavaScript in the browser console (10 minutes)

1. Open any web page in your browser. Press **F12**, and click the **Console** tab.
2. Type each line below and press **Enter**. After each line, look at the result.

```
console.log("Hello, world");
```

```
const college = "Mandsaur University";  
college
```

```
let count = 0;  
count = count + 1;  
count
```

```
const days = ["Mon", "Tue", "Wed"];  
days.push("Thu");  
days.length
```

```
const note = { id: 1, title: "Welcome", text: "Hello" };  
note.title
```

```
function greet(name) {  
  return "Hello, " + name;  
}  
greet("your name")
```

Check: You saw a result after each line, and you can say what a variable, an array, an object and a function are.

Close the tools with **F12**.

Part 7: Make the Notes Board work (35 minutes)

We now connect JavaScript to the page. Work in your `notes-board` folder.

7a. Update `index.html`

Open `index.html` and make five small changes.

Change 1. Under the `<textarea>` line, add a character counter:

```
<p class="counter"><span id="char-count">0</span> / 300 characters</p>
```

Change 2. Under the `<button>` line, add a place for messages:

```
<p id="form-message" class="message" role="status"></p>
```

Change 3. Change the heading `All notes` to show a number:

```
<h2>All notes <span id="note-count" class="badge"></span></h2>
```

Change 4. Delete the three sample notes. Replace the whole `<div id="notes-list">` block with an empty one. JavaScript will fill it:

```
<div id="notes-list"></div>
```

Change 5. At the very end of `<body>` , just before `</body>` , add the script tag:

```
<script src="js/app.js"></script>
```

7b. Create `js/app.js`

1. Click the `notes-board` folder, click **New Folder**, and name it `js` .
2. Click the `js` folder, click **New File**, and name it `app.js` .
3. Type the code below, **one section at a time**. Your instructor will explain each section before you type it.

Section 1: The data

```
// app.js (Day 2)
//
// The Notes Board now works in the browser.
// Our notes live in a JavaScript array. When you refresh the page,
// the array is created again and your new notes are lost.
// On Days 3 and 4 we fix this with a server and a database.

// 1. Data: a list (array) of notes. Each note is an object.
const notes = [
  { id: 1, title: "Welcome", text: "This is a sample note. Try adding your own." },
  { id: 2, title: "Study plan", text: "Learn HTML on Monday, CSS on Tuesday, and
JavaScript on Wednesday." }
];

let nextId = 3;
```

Section 2: Find the parts of the page

```
// 2. Find the parts of the page we need to change.
const form = document.getElementById("note-form");
const titleInput = document.getElementById("note-title");
const textInput = document.getElementById("note-text");
const charCount = document.getElementById("char-count");
const message = document.getElementById("form-message");
const notesList = document.getElementById("notes-list");
const noteCount = document.getElementById("note-count");
```

Section 3: Show a message

```
// 3. Show a short message under the form.
function showMessage(text, isError) {
  message.textContent = text;
  message.className = isError ? "message error" : "message ok";
}
```

Section 4: Draw the notes

```
// 4. Draw all notes on the page.
function renderNotes() {
  // Remove what is on the page now, then draw again from the array.
  notesList.textContent = "";

  if (notes.length === 0) {
    const empty = document.createElement("p");
    empty.className = "empty";
    empty.textContent = "No notes yet. Add your first note above.";
    notesList.appendChild(empty);
  }

  for (const note of notes) {
    const article = document.createElement("article");
    article.className = "note";

    const heading = document.createElement("h3");
    heading.textContent = note.title;

    const body = document.createElement("p");
    body.textContent = note.text;

    const deleteButton = document.createElement("button");
    deleteButton.type = "button";
    deleteButton.className = "delete-button";
    deleteButton.textContent = "Delete";
    deleteButton.addEventListener("click", function () {
      deleteNote(note.id);
    });

    article.appendChild(heading);
    article.appendChild(body);
    article.appendChild(deleteButton);
    notesList.appendChild(article);
  }

  noteCount.textContent = notes.length;
}
```

Section 5 and 6: Add and delete

```
// 5. Add a note to the array (newest note first).
function addNote(title, text) {
  const note = { id: nextId, title: title, text: text };
  nextId = nextId + 1;
  notes.unshift(note);
  renderNotes();
}

// 6. Remove a note from the array.
function deleteNote(id) {
  const index = notes.findIndex(function (note) {
    return note.id === id;
  });

  if (index !== -1) {
    notes.splice(index, 1);
  }

  renderNotes();
}
```

Section 7, 8 and 9: Events and the first draw

```
// 7. When the form is submitted, check the input and add the note.
form.addEventListener("submit", function (event) {
  // Stop the browser from reloading the page.
  event.preventDefault();

  const title = titleInput.value.trim();
  const text = textInput.value.trim();

  if (title === "" || text === "") {
    showMessage("Please write both a title and a note.", true);
    return;
  }

  addNote(title, text);
  form.reset();
  charCount.textContent = "0";
  showMessage("Note added.", false);
  titleInput.focus();
});

// 8. Show how many characters are used while the user types.
textInput.addEventListener("input", function () {
  charCount.textContent = textInput.value.length;
});

// 9. Draw the notes once when the page opens.
renderNotes();
```

7c. Add the new styles

Our new elements need style. Open the file `Day-2-Tools-and-JavaScript\done\notes-board\css\style.css`. Copy everything **after** the line `/* ----- Added on Day 2 ----- */` and paste it at the **end** of your `css\style.css`. Save.

7d. Test it

1. Open `index.html` in the browser (double-click it, or press **F5** if it is already open).
2. Press **F12** and look at the **Console** tab. It should show **no red errors**.
3. Try these tests:
 - Add a note with a title and text. It appears at the top.

- Press **Add note** with empty boxes. You see an error message.
- Click **Delete** on a note. It disappears, and the number changes.
- Type some text in the note box. The counter changes.
- Add a note with this text: `<b>bold</b>` . It shows as plain letters. This is the safe `textContent` working.
- Refresh the page with **F5**. All your new notes are gone. This is what we will fix on Days 3 and 4.

Check: Add, delete, error message and counter all work. If something fails, read the **Console** for a red message, and see the Stuck Sheet.

Part 8: Make the About page use JavaScript (15 minutes)

Today the details on your About page are typed into the HTML. We now keep them in one JavaScript file, so you only edit one small file.

8a. Create `js/student-details.js`

Create the file `js\student-details.js` and type this. Put **your own details** between the quotes. **You decide what to show.** Leave a value empty (`" "`) to hide it.

```
// student-details.js
//
// This is the ONLY file you need to edit to change the About page.
//
// How to use it:
// 1. Type your own details between the quotes.
// 2. You decide what to show. Leave a value empty ("" ) to hide it.
// 3. Save the file and refresh the About page.
//
// We encourage you to keep the instructor details at the bottom, so that
// your visitors know where this project was learned.

const studentDetails = {
  name: "", // Example: "Asha Sharma"
  rollNumber: "", // Example: "BCA2024-017"
  className: "BCA Second Year",
  college: "Mandsaur University",
  githubProfile: "", // Example: "https://github.com/your-username"
  tagline: "" // Example: "I love building small useful apps."
};

const instructorDetails = {
  name: "Rahul Dhangar",
  role: "Instructor",
  link: "https://github.com/rahuldhangar"
};
```

8b. Copy js/about.js

1. Open `Day-2-Tools-and-JavaScript\done\notes-board\js\about.js` .
2. Copy its whole content into a new file `js\about.js` in your own project.
3. Read it slowly. It uses the same ideas as `app.js` : it finds elements, creates new elements, and uses `textContent` . It also checks that a link starts with `https://` .

8c. Update about.html

1. In your `about.html` , find the `<section class="card">` that has the heading **Built by**. Replace the whole `<dl class="details"> ... </dl>` list and the hint paragraph under it with:

```

<!-- JavaScript fills this list from js/student-details.js -->
<dl id="student-details" class="details"></dl>
<p id="student-tagline" class="tagline"></p>

```

2. Find the last section (**Guided by**). Replace it with:

```

<section id="instructor-section" class="card">
  <h2>Guided by</h2>
  <!-- JavaScript fills this line from js/student-details.js -->
  <p id="instructor-line"></p>
</section>

```

3. At the very end of `<body>` , before `</body>` , add:

```

<script src="js/student-details.js"></script>
<script src="js/about.js"></script>

```

The order matters. `student-details.js` must come **before** `about.js` , because `about.js` uses the details.

4. Save, and open `about.html` in the browser.

Check: Your details appear. Empty values are hidden. The instructor line shows with a link. Now change one detail in `student-details.js` , save, refresh, and see it change on the page.

Part 9: Your first server (15 minutes)

1. In the `Workshop` folder (next to `notes-board`), create a new folder named `hello-server` .
2. Inside it, create a file named `hello.js` and type:

```
// hello.js
// Our very first server, written with Node.js only (no extra packages).
//
// Run it:  node hello.js
// Open it: http://localhost:3000
// Stop it: press Ctrl + C in the terminal

const http = require("node:http");

const server = http.createServer(function (request, response) {
  // This function runs every time a browser asks for a page.
  console.log("Request received for:", request.url);

  response.writeHead(200, { "Content-Type": "text/plain; charset=utf-8" });
  response.end("Hello from my first Node.js server!");
});

server.listen(3000, function () {
  console.log("Server is running. Open http://localhost:3000 in your browser.");
});
```

3. In the terminal, go to the folder and run it:

```
cd hello-server
node hello.js
```

4. The terminal prints `Server is running...` and **stays busy**. This is correct. The server is waiting for requests.
5. Open your browser and go to `http://localhost:3000`. You see `Hello from my first Node.js server!`.
6. Look at the terminal. It printed `Request received for: /`. Your browser **requested**, and the server **responded**.
7. In the browser, try `http://localhost:3000/about`. The terminal prints a different path, but the answer is the same.
8. Go back to the terminal and press **Ctrl + C** to stop the server.

Check: You can explain in your own words: "The browser sent a request. The function in `hello.js` ran. The server sent a response."

Part 10: Backup, quiz and homework

1. Copy your `Workshop` folder to a USB drive or to the place your instructor tells you.
 2. Do the 5-question quiz that your instructor shares.
 3. Read the **Homework for Day 3** at the end of the Day 2 Theory.
-

Final checklist

`node -v` shows version 22, and `node setup-check.js` shows four PASS lines.

My Notes Board adds and deletes notes, and shows an error for empty input.

My About page shows the details I chose, and the instructor line.

I ran `hello.js` and opened it in the browser.

I made a backup copy of my `Workshop` folder.

Extra challenges (only if you finish early)

1. Show the date and time each note was created. Hint: `new Date().toLocaleString()`.
2. Add a button **Clear all notes** that empties the array.
3. Change `hello.js` so that the server says your name.
4. Add a `tagline` to your `student-details.js` and see it on the About page.
5. In the Console of your Notes Board page, type `notes` and press Enter. What do you see? Why?